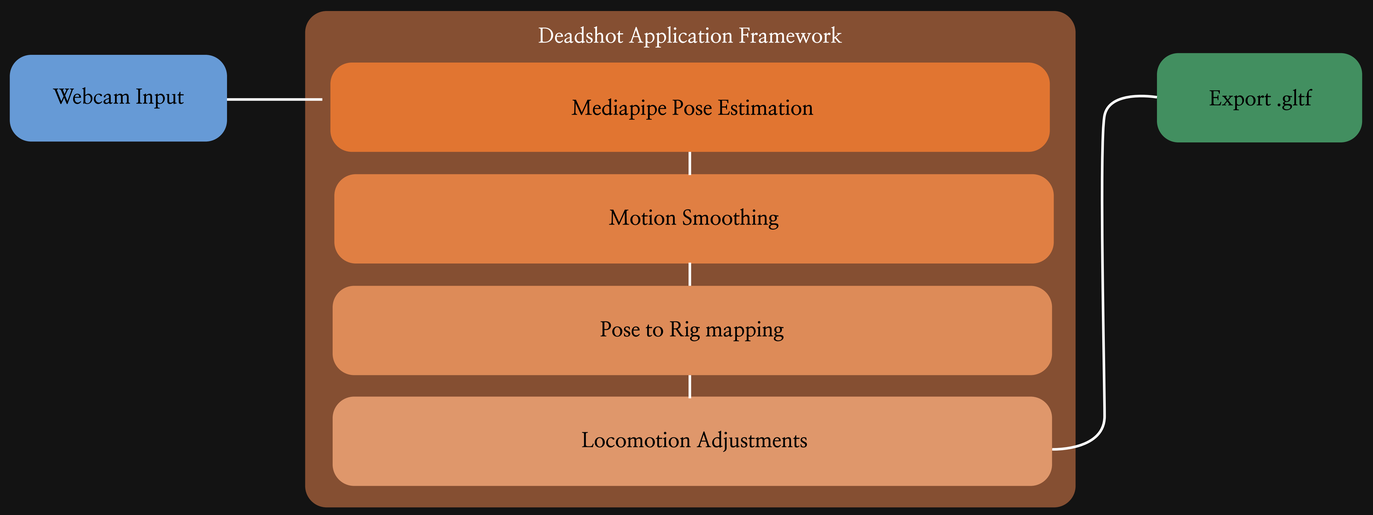
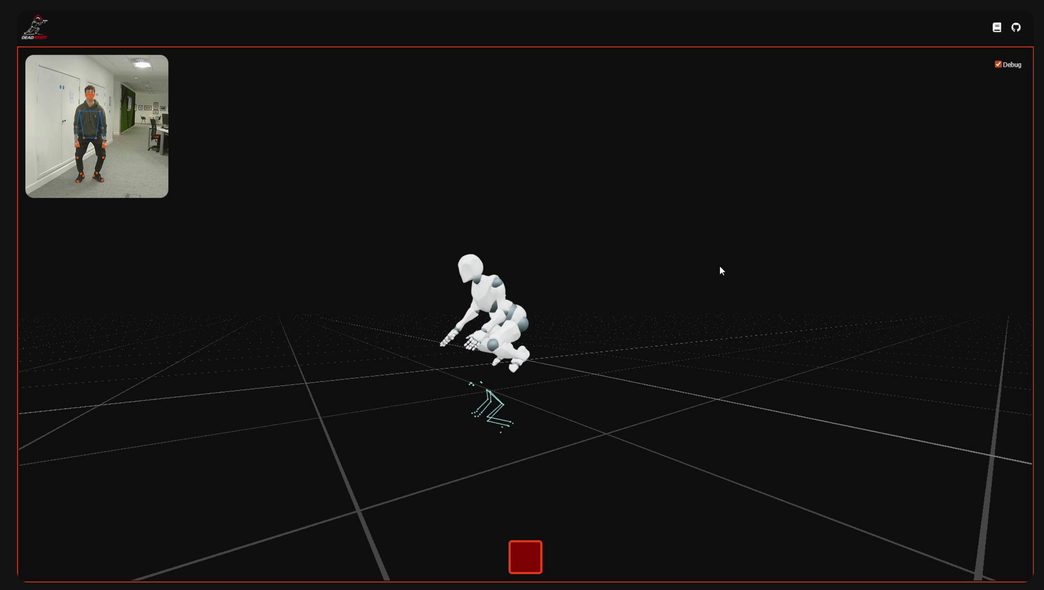
# Deadshot: A Web-Based, Open-Source Framework for Accessible Motion Capture

Christopher Hosken (s5605094)
BA Computer Animation Technical Arts

## PIPELINE



## OVERVIEW

We present Deadshot, an open-source, browser-based framework for accessible monocular motion capture. While video-based pose estimation offers a low-cost alternative to conventional techniques, costly closed-source codebases, technical friction in offline setups, and insufficient motion data accuracy remain significant barriers to public accessibility and adoption within the Computer Graphics industry. Deadshot explores these challenges by integrating existing, open-source pose-estimation frameworks with novel algorithms for temporal-velocity-confidence smoothing and joint retargeting. This work demonstrates a complete, browser-based system, serving as both a technical offering and study in the modern world of motion capture technology.

## METHODOLOGY

Deadshot utilizes the open-source Mediapipe framework for monocular pose estimation, which outputs a skeleton containing 2D screen coordinates (x, y), an estimated depth coordinate (z), and a prediction confidence (c) for each joint.

To reduce jitter in the post-estimation, a proprietary pre-processing stage known as adaptive velocity-confidence smoothing is introduced before retargeting and locomotion adjustments.

A Quaternion-based joint retargeting technique ensures that the character's movement angles remain consistent, regardless of any scaling issues caused by z-depth inaccuracies.

To compensate for the loss of all translational data inherent in joint-rotation mapping, the framework implements a continuous live simulation of the pelvis.

For further usage in Digital Content Creation, recoded motion sequences are exported using the popular .gtlf format.

## SMOOTHING ALGORITHM

A novel adaptive velocity-confidence smoothing algorithm is applied to estimated points for smoothing.

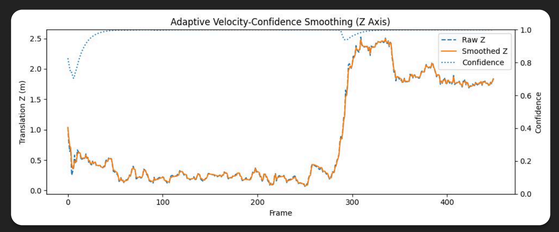
$$v_i = \|\vec{P}_n - \vec{P}_{n-1}\| = \sqrt{(x_n - x_{n-1})^2 + (y_n - y_{n-1})^2 + (z_n - z_{n-1})^2}$$

$$\alpha_{\text{base}} = \begin{cases} \alpha_{\text{slow}}, & v_i < T \\ \alpha_{\text{fast}}, & v_i \geq T \end{cases}$$

$$\alpha_{final} = \alpha_{base} \cdot \text{clamp}(c, 0.2, 1.0)$$

$$S_n = S_{n-1} + \alpha_{final} \cdot (\vec{P}_n - S_{n-1})$$

25.17% reduction in high-frequency jitter with a 0.15% reduction in overall jitter.




## TECHNICAL TOOLS

The Deadshot framework is built as a web application using Vite+React Typescript and THREE.js as the 3D engine, selected for its cross-platform compatibility and ease of development.

## ACKNOLWEDGEMENTS

I would like to thank my unit leader Valery Adzhiev and project supervisor Zhidong Xiao for their critiques and contributions to this project.

## CONCLUSION & CRITICAL ANALYSIS

The framework is highly accessible, user-friendly, and demonstrates a clear path for exporting motion data. The main limitation is the quality of the underlying pose-estimation model z-depth data.

Future work will focus on three key areas:

- Training a new web-compatible pose-estimation model with higher z-depth accuracy.
- Porting more accurate, locally run 3D pose-estimation technique to the web.
- Implementing custom WASM bindings of FBX or USD for broader compatibility with DCCs.

## LIVE DEMO

## REFERENCES

Google. (2024, June 7). google-ai-edge/mediapipe. GitHub. https://github.com/google-ai-edge/mediapipe
Google. (2025, January 13). Pose landmark detection guide | Google AI Edge. Google for Developers. https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker
Jocher, G. (2020, May 18). YOLOv8 Documentation. Docs.ultralytics.com. https://docs.ultralytics.com/
Move AI. (2026, February 11). MOVE Ai. Www.move.ai. https://www.move.ai/
Nogueira, P. (2011). Motion Capture Fundamentals a Critical and Comparative Analysis on Real-World Applications. https://paginas.fe.up.pt/~prodei/dsie12/papers/paper_7.pdf
RadicalMotion. (2026, February 11). RADiCAL - AI-powered 3D animation. RADiCAL. https://radicalmotion.com/
Riki, J. K. (2013, November 18). How Fast Should You Animate? Animator Island. https://www.animatorisland.com/how-fast-should-you-animate/
Rokoko. (2025, December 13). Rokoko - Animation and MoCap Tools for all Creators. Rokoko.com. https://www.rokoko.com/
Wandt, B., Ackermann, H., & Rosenhahn, B. (2026). 3D Human Motion Capture From Monocular Image Sequences. Cv-Foundation.org, 1(1), 1-8. https://www.cv-foundation.org/openaccess/content_cvpr_workshops_2015/W06/html/Wandt_3D_Human_Motion_2015_CVPR_paper.html
Xsens. (2025, March 10). Xsens Motion Capture Pricing? Xsens.com; Movella. https://www.xsens.com/resources/xsens-motion-capture-pricing